Department of Chemical Engineering, IIT Kharagpur
End-Semester Examination, 2013
Transport Phenomena (CH30012)
Open Book, Closed Notes Examination, Photocopies/Sharing of Books Not Allowed

The following textbooks are allowed i) Fox & McDonald ii) Bird Stewart & Lightfoot iii) Incropera & Dewitt

1. Consider a system of two concentric rotating cylinders. The two cylinders each rotate at a constant but different angular velocity and you may neglect body forces.
   a) Determine the velocity profile $v_\theta(r)$ between the cylinders and the pressure distribution P(r).
   b) Determine a friction factor by calculating the force required to turn either of the two cylinders. Given that for the outer cylinder, one can express the force and the friction factor as (v_o is the linear velocity of the outer cylinder): **10**

$$F = \tau_{r\theta} A = C_f\left(\frac{1}{2}\rho v_o^2\right)A$$

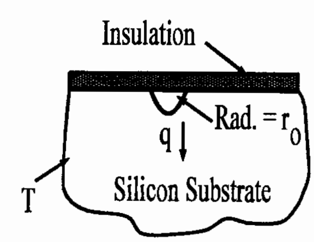
2. In your haste to get to the shopping mall you forgot to take your heat sensitive medicine with you and left it in the car. The interior dimensions are (L = 2 m, Area = 2.5 m^2, Volume = 4 m^3). Solar radiation provides a heat flux of 116.5 W/m^2 through the sun-roof and windows (Area = 2.5 m^2). A breeze (v_∞ = 5 m/s, T_∞ = 25 °C) blows over the car. From previous experiments, you know that the friction factor for flow over the car obeys:

$$\frac{\overline{c_f}}{2} = 0.05\mathrm{Re}_l^{-0.35}$$

   Assume only air inside the car ($v_{air} = 0$; T(t = 0) = 25 °C), the fact that the costly and sensitive medicine can withstand a maximum temperature of 55 °C, and that all the net energy from the sun gets transferred to the air inside the greenhouse, and that the temperature variation of air inside the car can be neglected:
   a) What is the heat transfer coefficient?
   b) How fast does the temperature initially rise or fall in the greenhouse?
   c) Will the medicine remain effective before you have finished your shopping? Why?
   Given: $\gamma = 15.7\times10^{-6}$ m^2/s, Pr ≈ 1, k_f = 0.026 W/m K, ρ=1.18 kg/m^3, c_p=1005 J/kg K **10**

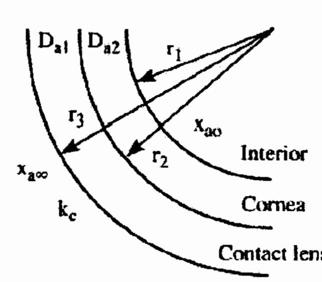
3. Slug flow is an idealized tube flow condition for which the velocity is assumed to be uniform over the entire tube cross section. For laminar slug flow with a uniform surface heat flux, determine the form of the fully developed temperature distribution, T(r) and the Nusselt number starting with the energy equation. **8**

4. A transistor, which may be approximated as a hemispherical heat source of radius r_o = 0.1mm, is embedded in a large silicon substrate (k = 125 W/m.K) and dissipates heat at a rate q. All boundaries of the silicon are maintained at an ambient temperature of T_∞ = 27°C, except for a plane surface which is well insulated. Obtain a general expression for the substrate temperature distribution in terms of the ambient temperature (T_∞) and the surface temperature of the heat source. Find the surface temperature of the heat source for q = 4W. **7**



5. Benzene, a harmful chemical, has been spilled on the laboratory floor and has spread to a length of 2m. If a film 1 mm depth is formed, how long will it take for the Benzene to completely evaporate? Ventilation in the laboratory provides for airflow parallel to the surface at 1 m/s, and the benzene and air are both at 25 C. The mass densities of Benzene in the saturated vapor and liquid states are known to be 0.417 and 900 kg/m^3, respectively. For Air: kinematic viscosity = 15.7x10^{-6} m^2/s; D_{AB} = 0.88x10^{-5} m^2/s, Sc = 1.78. **8**

6. Oxygen transfer from the atmosphere to the interior regions of the eye depends enormously on whether one wears contact lenses. Treating the eye as a composite spherical system, determine the mass transfer rate from the atmosphere through the cornea with and without contact lenses. You may assume dilute solutions. The oxygen concentration in the interior is maintained at a uniform level by circulation of the fluid inside the eye. **7**



Data: $x_{a\infty} = 0.2$ $\quad$ $x_{ao} = 0.05$ $\quad$ $r_1 = 0.0102$ m
$r_2 = 0.0127$ m $\quad$ $r_3 = 0.0165$ m $\quad$ $D_{a1} = 1\times10^{-9}$ m^2/s
$D_{a2} = 5\times10^{-9}$ m^2/s $k_c = 1\times10^{-3}$ m/s^{-1} $\quad$ $c_t = 0.008$ moles/m^3